

One interface, four front ends

Each tinted box is one drive; the two left panels of a box are two different devices on that car. Badge “paper” marks the front end used in the paper’s experiments, “demo” the second device shown to make the interface point.

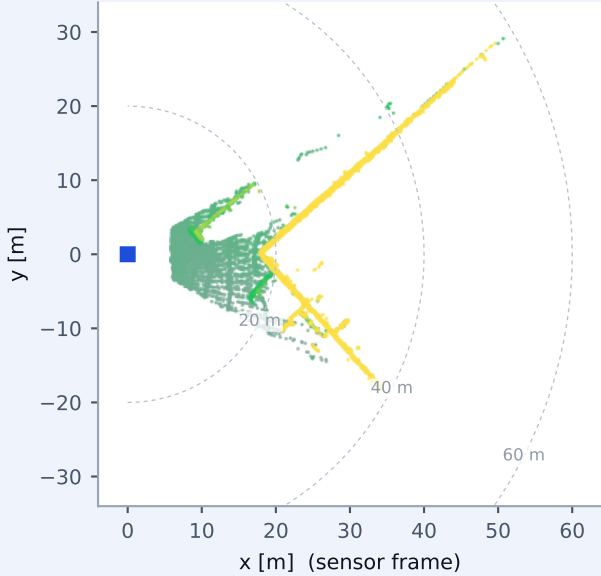
A2D2 · drive 20180810_150607

two independent devices · nothing shared: no field of view, no clock, no measured quantity

SENSOR A · LiDAR

paper

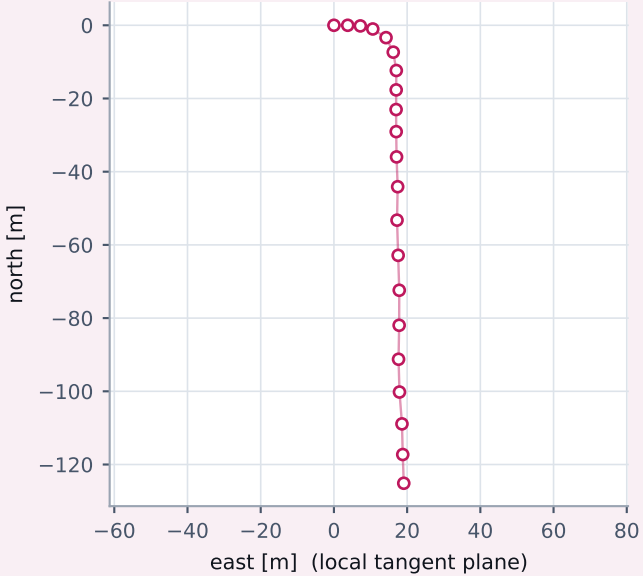
Velodyne VLP-16, FRONT_CENTER view · ≈ 30 Hz
range + bearing to surfaces → scan matching
shown: one scan, 14 000 points



SENSOR B · GNSS

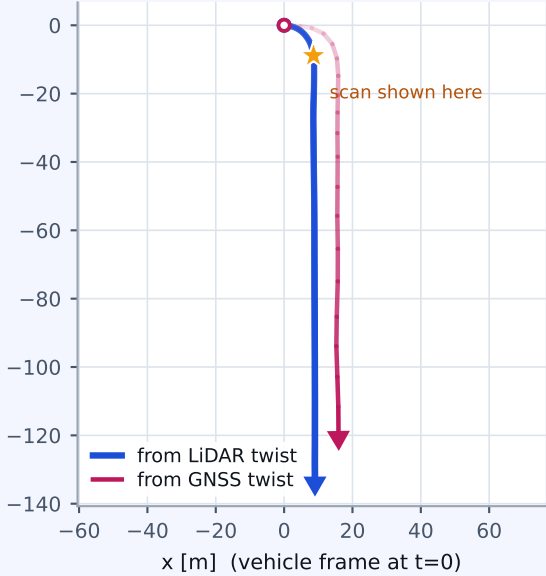
demo

vehicle-bus GNSS, position only · 1 Hz fix
absolute position on Earth → differencing fixes
shown: 21 fixes at 1.0 Hz, no attitude



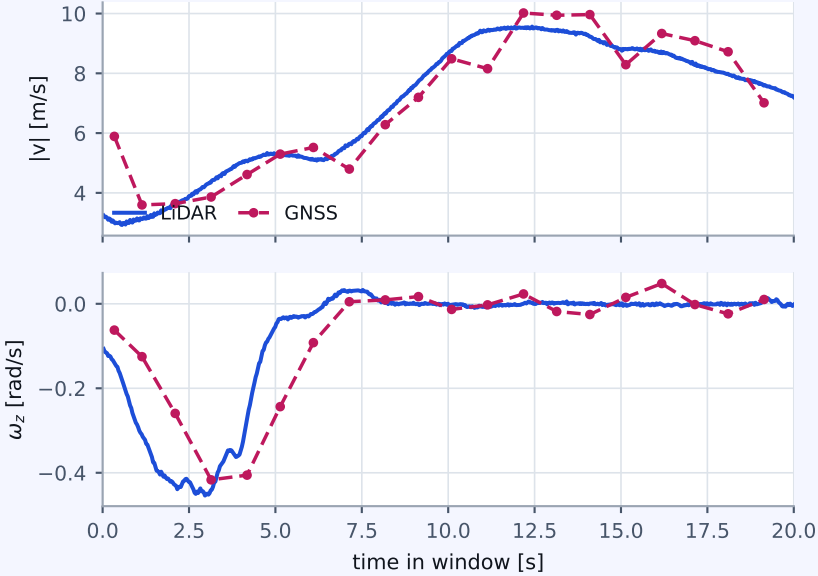
Trajectory from (v, ω)

139 m driven, -88° of heading



The twist, as delivered

path speed 6.95 vs 6.98 m/s · median ω gap 0.0242 rad/s



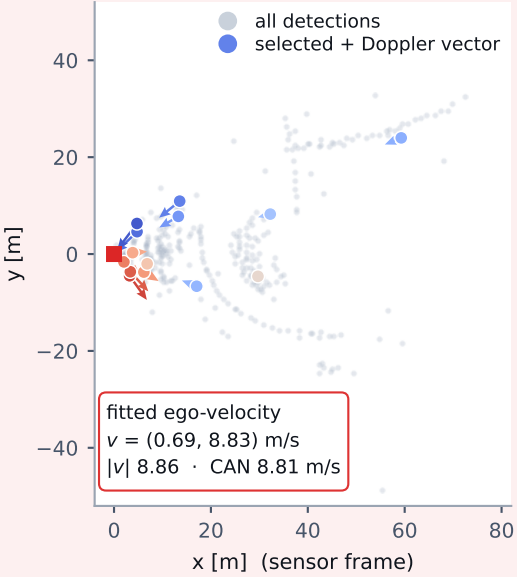
RadarScenes · sequence 1

two independent devices · nothing shared: no field of view, no clock, no measured quantity

SENSOR A · radar

paper

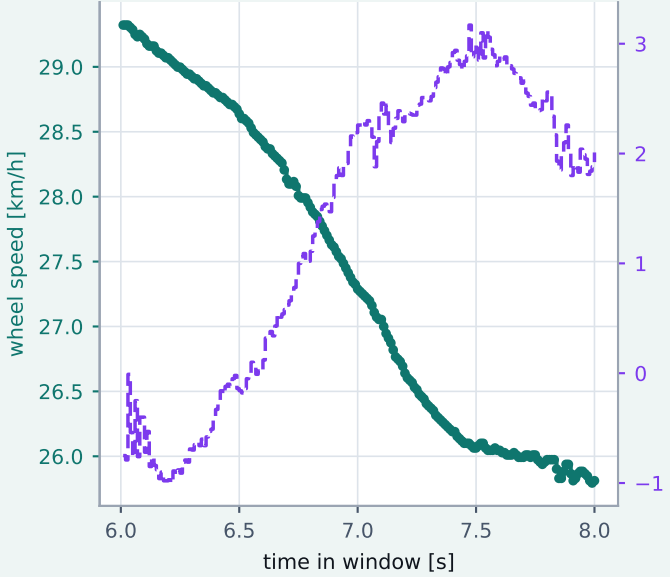
77 GHz series radar, sensor 1 · 13.5 scans/s
range, azimuth, Doppler → one-scan least squares
shown: one scan, 310 returns · 14 vectors



SENSOR B · CAN bus

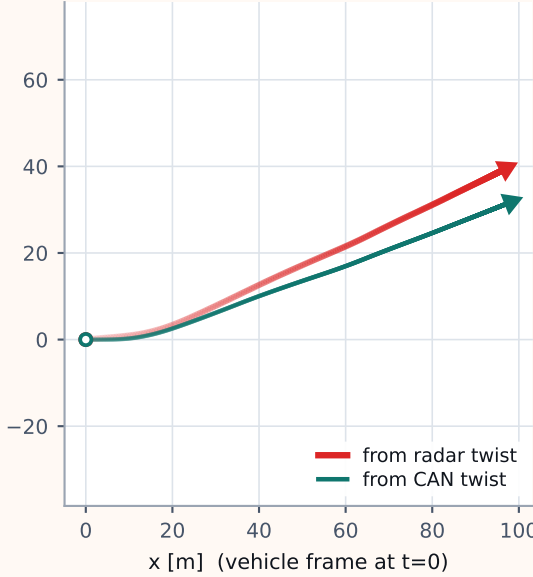
demo

wheel encoders + yaw-rate gyro · 100 Hz
wheel rotation + yaw rate → vehicle kinematics
shown: 2 s zoom of the window yaw rate [deg/s]



Trajectory from (v, ω)

105 m driven, 22° of heading



The twist, as delivered

median $|v|$ gap 0.078 m/s · one 2-D radar gives no ω

